

Omdena HYD

***LEVERAGING LLMS TO
UNDERSTAND
GLOBAL MENTAL HEALTH
WELL-BEING & FOMO***

TEAM - 5

FINAL DEMO

CONTENTS

01

Problem
Statement

02

Goals and
Objective

03

Datasets

04

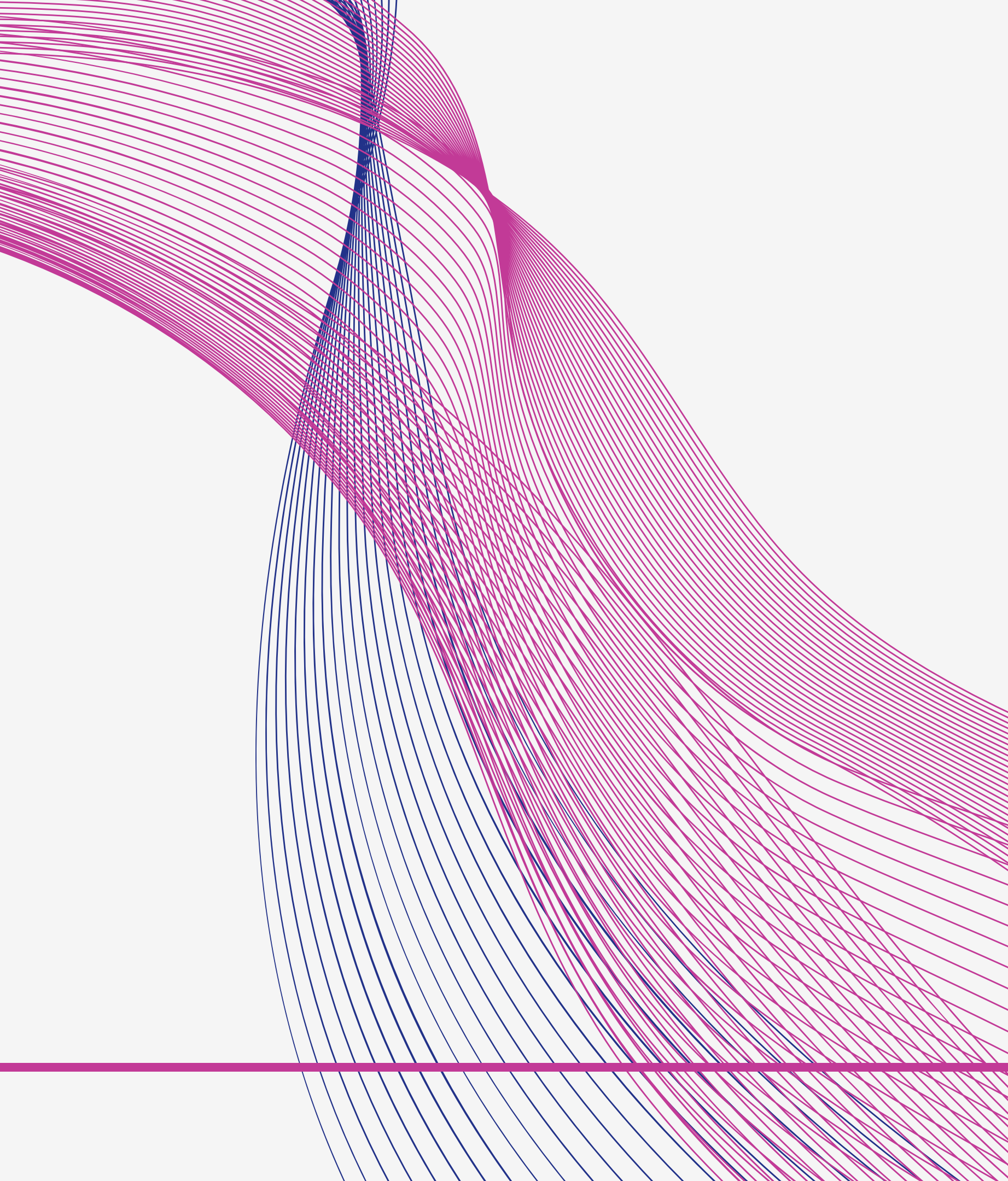
Model
Building

05

Prototype
Demo

06

Future Works
and
Improvements



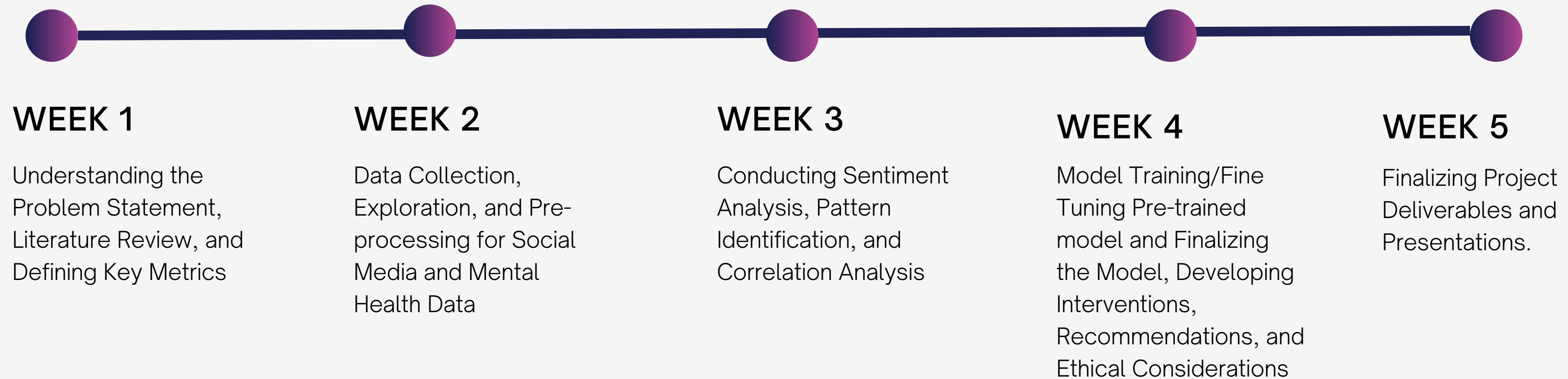
PROBLEM STATEMENT

Social media platforms have become ubiquitous, fundamentally altering how we connect and interact. This pervasive influence necessitates a closer look at their impact on mental health. While social media fosters connection, self-expression, and community building, concerns regarding its potential to exacerbate anxiety, depression, and feelings of inadequacy are rising.

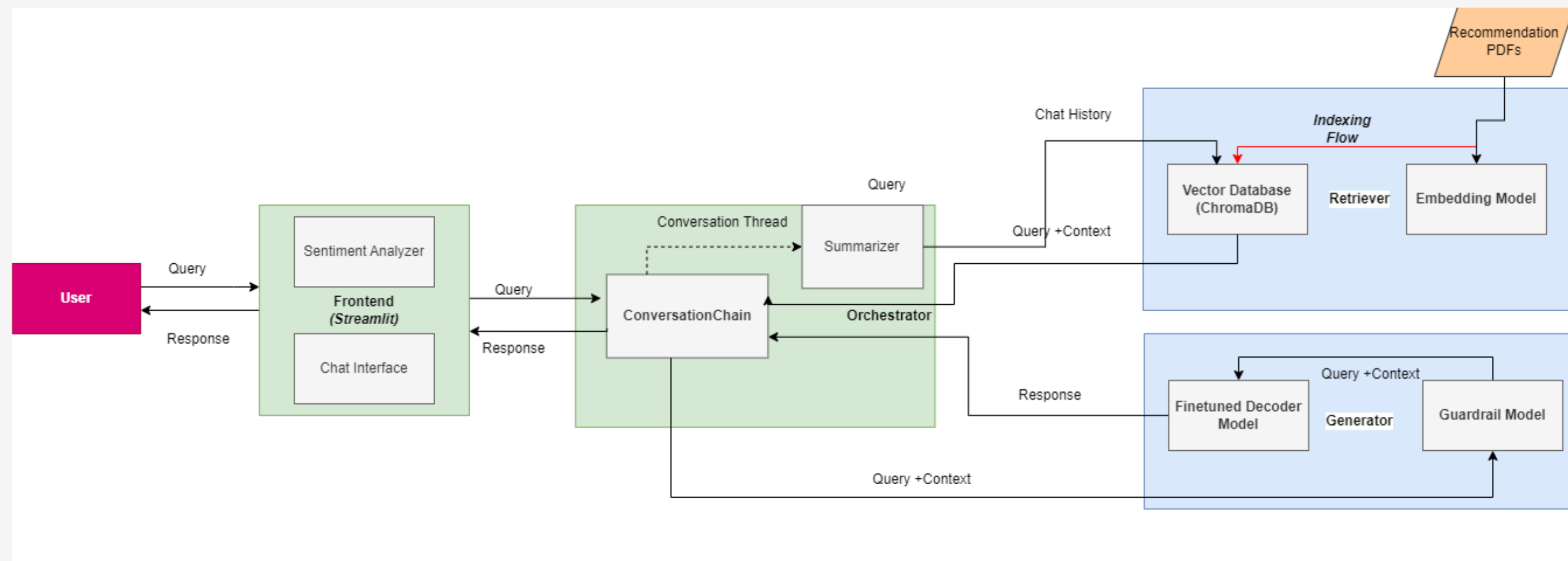
Goals and OBJECTIVES

- Our aim is to pioneer a chatbot design to tackle prevalent challenges in mental health support.
- The primary goal is to offer users a space to articulate their emotions and concerns securely. Through advanced analysis of user inputs, the chatbot endeavors to detect potential signs of mental health issues.
- Additionally, it will deliver tailored interventions and provide access to relevant recommendations.
- Emphasizing empathetic and responsive design, our objective is to cultivate positive interactions and sustain user engagement.

PROJECT TIMELINE



CHATBOT DESIGN



Data Collection

Two types of data:

1. **Primary Data**- This is the data that is scraped by the team from Quora, Facebook, Reddit posts.
2. **Secondary Data**- Already existing datasets from Kaggle, Research Papers etc.

Data Preprocessing

Data cleaning and Preprocessing Steps:

- Removing Unnecessary Columns
- Text Cleaning
- Lowercasing
- Removing Special Characters
- Handling Missing Values.
- Ensuring Uniqueness

DATASETS

01

Conversational Data

- i. **98 QnA pairs of Frequently Asked Questions**
- ii. **Counsellor_chats: 824 Unique Questions**
- iii. **98 Q&A pairs of Human and Therapist Conversations from mental health chatbot collected from Kaggle.**
- iv. **100 Q&A Pairs from data scrapped from Facebook**

03

Recommendation/ Intervention Strategies

- i. **Parents Guide**
- ii. **Depression guides from helpguide.org**

02

Demographical Data

This dataset belongs to a sample of 561 university students from both public and private universities in Malaysia. This dataset consists of the demographic characteristics of the sample and the data on two main variables, cyberbullying victimisation and the negative emotional states.

04

Classification Data without features

- i. **20000 rows of texts with classification found in Kaggle scrapped from Reddit .**
- ii. **26923 rows of text classification data source from Kaggle.**
- iii. **7000+ rows of text classification data source from Twitter**

MODEL BUILDING

Encoder - only MODELS

- We used DistilBERT for sequence classification task
- We are classifying user queries into classes:
Depressed or Non-Depressed
- We used the Depression_dataset_preprocessed dataset, which is balanced.
- Model evaluation results show F1, Precision, and Recall all above 90%



Decoder Only MODELS

- We used Mistral 7B Instruct v2.0 for the chatbot model.
- We trained with the SFT Trainer using the following hyperparameters:
Epochs: 2, Optimizer: Adam, Learning Rate: 0.0004.
- Then, we trained with the DPO Trainer using the following hyperparameters:
Epochs: 5, Optimizer: Adam, Learning Rate: 0.0002.
- In human evaluation, the finetuned model is working well.

RESPONSIBLE AI MODEL



- We utilized the **LLAMA Guard** model for responsible AI
- By integrating LLAMA Guard, we uphold ethical standards and foster a safe and trustworthy environment for our users.
- LLAMA Guard's safety taxonomy includes categories such as Violence & Hate, Sexual Content, Guns & Illegal Weapons, Regulated Substances, Suicide & Self Harm, and Criminal Planning.
- We used LLAMA Guard classification model to categorize user prompts into two categories: safe and unsafe. If a prompt is deemed unsafe, the chatbot will not respond; otherwise, it will engage in normal conversation.

Model DEPLOYMENT

Deployment Platform:

- Utilizing Hugging Face's user-friendly platform for streamlined model deployment.

Hardware Utilization:

- Free CPU Basic: The foundational deployment is supported by Hugging Face's basic free hardware, ensuring continuous availability of the chatbot for users.
- Dedicated Instance: For enhanced performance, a dedicated, more powerful instance is employed specifically for the LlamaGuard feature.

LlamaGuard Feature:

- On-Demand Activation: LlamaGuard, a pivotal feature ensuring user privacy and security, is activated on-demand, providing users with peace of mind during interactions.

Demo of PROTOTYPE

Recommendation

Here are some suggestions to help you feel better when you're feeling sad, alone, and dealing with a recent breakup:

1. Volunteer your time to help others.
Helping others is a great way to boost your mood and expand your social network.
2. Join a depression support group.
Connecting with others who are going through similar experiences can help reduce your sense of isolation and provide valuable insights and coping strategies.
3. Talk to someone you trust. Sharing your thoughts and feelings with someone who listens and understands can help you feel less alone and more supported.
4. Consider reaching out to a mental health professional for additional support and guidance.

Mental Health Counseling Chatbot

Deploy

 I am feeling sad

 I'm really sorry that you're feeling this way, would you like to share what's making you feel sad?

 I am your Personal Therapist, How are you doing today?

 i had a breakup recently

 I'm really sorry to hear that you're going through a tough time right now. Could you tell me a bit more about what's been happening? Maybe share some details about the relationship and what led to the breakup? This might help us understand things better.

Please tell me about your mental health condition and we can explore together potential advice that could help you.



FUTURE WORKS AND IMPROVEMENTS

- Replace LLM with Fine Tuned LLM
- Deploy a fully functional health app linked with patients' medical history
- Model to better predict the Pos/Neg sentiment
- Model to recognise end of conversation
- Understand User Pattern
- Language Translation
- Advanced AI Features
 - LLMs and Multimodal Techniques
 - Predictive Modeling

THE TEAM

Krishna Kishore Gajjala

Ankitha Beeram

Debanjan Mondal

Phaneendra Ganji

Victoria Voloshyna

Elizabeth Waithaka

Meriem Baziz

Rachel Shirin Padibandla

Sayak Mallick

Vijaya Krishna

Dhruv Tibarewal

Ky Nguyễn

Obinna Ekesi

Amos Fang

Pranav Gavali

Reem Abdel-Salam

Team Reflections

Victoria Voloshyna

I am really grateful for the chance to participate in the project. I have learned a lot of new skills on building chatbots and brushed up my old analytical skills. Thanks for being together on this journey, team.

Dhruv Tibarewal

I extend my heartfelt thanks for the invaluable opportunity to contribute and collaborate on a real-life project focused on global mental health and FOMO using LLMs

Obinna Ekesi

What inspired me most is the drive to get the job done as a team; teamwork was topnotch. Appreciation goes to Krishna Kishore Gajjala for spectacular leadership qualities shown and every other team member for cooperation.

Ankitha Beeram

Reflecting on this journey, I am grateful for the invaluable learning experience it brought. It was a fantastic opportunity to obtain hands-on experience and the team was incredible to work with. I'm excited for the future, thanks to this experience and the connections I made.

Phaneendra Ganji

Really great full to work with such hard working team, everyone coordination is the result of the final product. I was completely new to the LLM, RAG space but i was able to learn everything from scratch starting with the what it is, how it is implemented and out of several ways to implement, found better ways from a great experimentation. Self learning on such regularly updating tech is hard but I enjoyed it.

Elizabeth Waithaka

I have thoroughly enjoyed working on this project. The hardwork really paid off and I am proud of what we have built as a team. I have learnt a lot from everyone who was a part of this project. I would also like to give a special thanks to Krishna Kishore Gajjala for his dedication in ensuring we got to the finish line.. Well done!

Debanjan Mondal

The Last Month On The Omdena Project Was Great! Every Day, I Looked Forward To Our Team Meetings. It Was Fun To Work With Such A Nice And Helpful Group. I Learned A Lot About Different Things, Like How Projects Are Planned (Rag Pipeline), How To Design User Interfaces (Ui), And How To Make Sure Ai Is Used Responsibly. There Were Some Tough Parts Too, But I Figured Them Out And Learned From Them. I'm Really Thankful For This Chance To Learn And Work With Others.

Sayak Mallick

I had an excellent experience working with motivated folks on an interesting real world application. picked up many new skills and knowledge on recent advancements related to llm's. Special thanks to Krishna Kishore Gajjala Debanjan Mondal, Ankitha Beeram for their constant support and camaraderie.

Vijaya Krishna

Great project, I learned lots of new things and saw the strongest team ever . Thanks to all the team members who supported me.

Team Reflections

Team 5

THANK YOU

